Standby Magnetic Compass – General Data

1. General

The vertical card magnetic compass (PAI–700) is used to indicate relative bearing to the earths magnetic poles.

2. Description

A. Vertical Card Magnetic Compass (PAI–700)

Unlike a “wet” compass the PAI–700 utilizes eddy current magnetic damping, no fluid is inside the instrument. The rotation of the sensing magnet is transferred to the azimuth compass card which indicates the relative bearing of the aircraft. Instrument lighting is controlled by the INSTRUMENT PANEL dim potentiometer.

3. Adjustment/Test

A. Function Test

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Start both engines. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

3) Wait several minutes for the aircraft electrical and avionics systems to initialize.

4) Move the prop levers to the cruise prop rpm position, (full aft) not feathered.

5) All in flight systems should be turned on.

6) Rotate the aircraft to a known magnetic NORTH heading. Note the degrees of deviation.

7) Rotate the aircraft to a known magnetic SOUTH heading. Note the degrees of deviation.

8) Verify the NORTH and SOUTH degrees of deviation is the same. If the degrees of deviation are not the same, do a compensation procedure, refer to Para 3.B.

9) Rotate the aircraft to a known magnetic EAST heading. Note the degrees of deviation.

10) Rotate the aircraft to a known magnetic WEST heading. Note the degrees of deviation.

11) Verify the EAST and WEST degrees of deviation is the same. If the degrees of deviation are not the same, do a compensation procedure, refer to Para 3.B.

12) Turn off all in flight systems that were turned on for the test.

13) Turn the engines OFF. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

14) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Error Compensation

CAUTION **USE A NON-MAGNETIC SCREW DRIVER FOR ADJUSTMENTS.**

Note Instrument friction during the heading checks can be minimized by tapping the top right hand portion of the case three to four times with a wooden pencil at each heading or by slightly tapping the upper right hand portion of the case with your forefinger.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Start both engines. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

3) Wait several minutes for the aircraft electrical and avionics systems to initialize.

4) Move the prop levers to the cruise prop rpm position, (full aft) not feathered.

5) All in flight systems should be turned on.

6) Position the aircraft to a known magnetic NORTH heading.

7) Use the N-S (LH screw) adjustment screw to remove all deviation. Verify the compass indicates N.

8) Rotate the aircraft to a known magnetic EAST heading.

9) Use the E-W (RH screw) adjustment screw to remove all deviation. Verify the compass indicates E.

10) Rotate the aircraft to a known magnetic SOUTH heading. Note the degrees of deviation.

11) Remove half of the total deviation from the SOUTH heading.

12) Rotate the aircraft to a known magnetic WEST heading. Note the degrees of deviation.

13) Remove half of the total deviation from the WEST heading.

14) Rotate the aircraft around the compass card (0 deg. to 330 deg.) at every 30 degree increment. Note the deviation.

15) Record the 30 degree deviation error on a compass correction card. Maximum tolerance of the compass deviation is 10 degrees.

Note Additional maintenance and inspection requirements for the installation of the Magnetic Compass may be mandated by the regulations of the jurisdiction under which the aircraft is registered. In the event of disagreement between the PSM 1-64-2 and these additional requirements, the more restrictive requirements shall take precedence unless an alternative program of maintenance has been approved by the regulatory authority.

16) Turn off all in flight systems that were turned on for the test.

17) Turn the engines off. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

18) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal and Installation

A. Removal and Installation Vertical Card Magnetic Compass (PAI–700)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove the screws (4) securing the compass unit to the windshield center post cover.

c) Disconnect the molex lighting connector.

d) Remove compass from aircraft.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Connect the molex lighting connector.

c) Install the screws (4) securing the compass unit to the windshield center post cover.

d) Do a function test, refer to Para 3.A.

5. Troubleshooting

A. Vertical Card Magnetic Compass (PAI–700)

1) Visually inspect the electrical connectors for damaged wiring or contacts. Ensure connectors are fully seated and tight.

2) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

3) If repair or service is required, send to appropriately rated avionics repair facility.

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